Sampling and Variability

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

Week 2: sampling can produce different outcomes from the same distribution.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Sample from a distribution using explicit cumulative intervals. Try the recall before opening notes.

Your learning target

I can sample from a distribution using explicit **cumulative interval**s.

I can calculate **temperature**-adjusted probabilities for a toy example.

I can explain why random variation and factual reliability need separate evaluation.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a **score** for your ability.

Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 4: **Sampling** and **Variability**

Decoding choices trade consistency, diversity, and risk.



CREATOR MISSION

Compare low- and high-**variability** story continuations.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Does the largest next-token probability guarantee the largest complete-path probability?

Readiness for A small attention calculation

What must happen to permitted weights after excluding a position?

My readiness explanation

This week’s end goal

By the end, I can calculate a toy **temperature** change and trace a **sampling** draw.

Notice the evidence

> Investigate the supplied Sampling and Variability case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **sampling** and **variability** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Start with probabilities red 0.6, blue 0.3, green 0.1 in that fixed order. Greedy chooses red. **Sampling** uses a draw u with 0 <= u < 1 and **cumulative interval**s: red [0,0.6), blue [0.6,0.9), green [0.9,1).

Left endpoints are included, right endpoints excluded. Thus u=0.6 selects blue and u=0.9 selects green. The draw 1 is outside this contract.

Sampling does not mean choosing all tokens equally. Across many independent draws the frequencies may approach the specified probabilities, but a small sample need not match them exactly.

Fixed recorded draws allow a reproducible classroom trace. A random seed and generator version help replay a real **sampling** program, not establish the truth of **output**s.

Apply the connected idea: A small attention calculation

For one query q and keys k_i of width d, compute **score** s_i = dot(q,k_i)/sqrt(d). Then obtain weights from **softmax** and combine value vectors. Dot products may be negative; probabilities cannot.

The **softmax** performs the conversion. This is a single-head classroom calculation, not a full language model.

Define **softmax** weights as exp(s_i-m)/sum_j exp(s_j-m), where m is the maximum permitted **score**.

Subtracting the same maximum from every permitted **score** leaves the ratios unchanged and avoids unnecessary overflow from large positive **score**s.

Natural exponent exp is introduced here; use a calculator or Python math.exp rather than estimating its meaning from the word exponential.

STUDENT LESSON · 3 OF 18

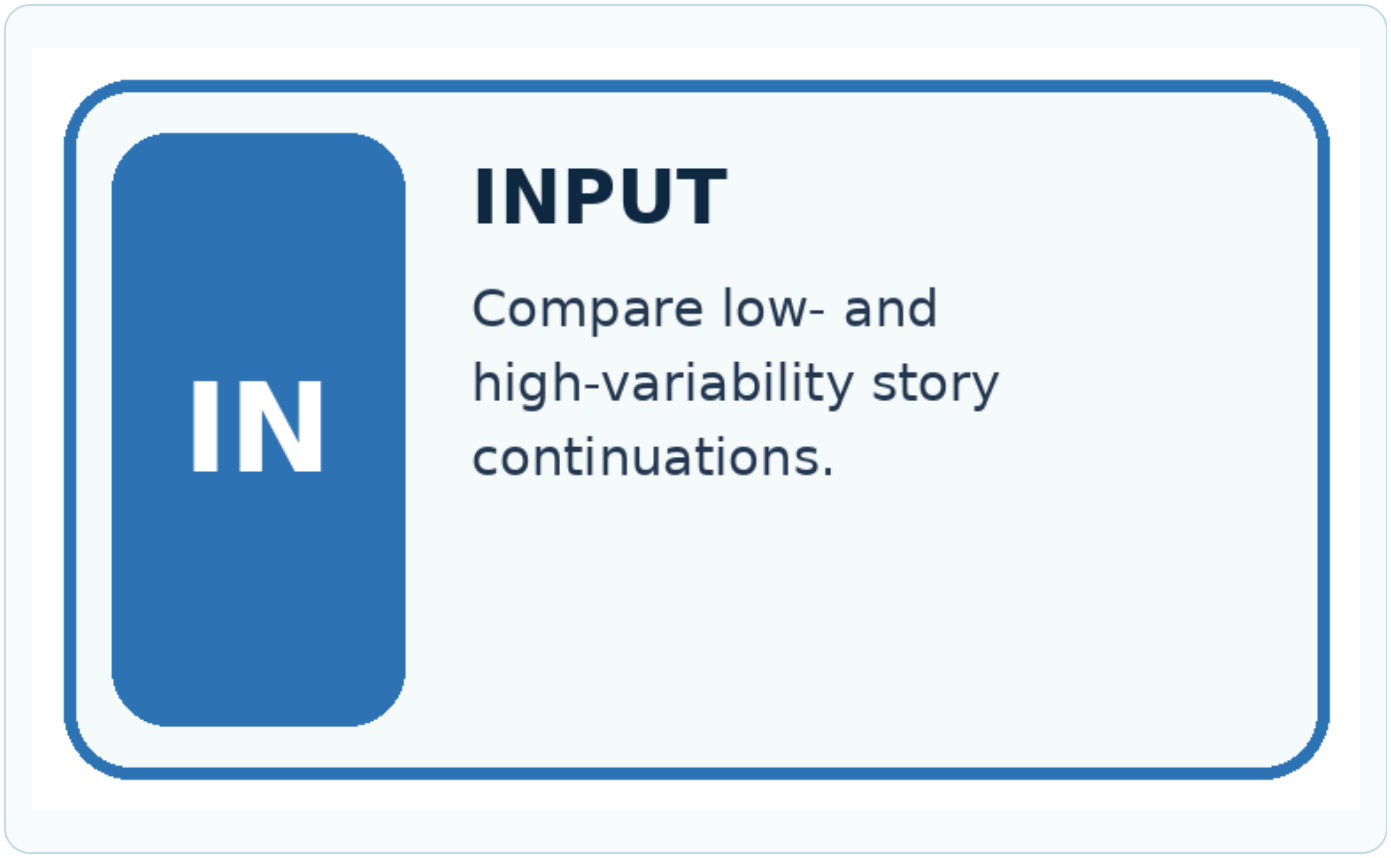
See how it works

> Follow the mechanism for Sampling and Variability. Connect each labelled input, deciding step and result.

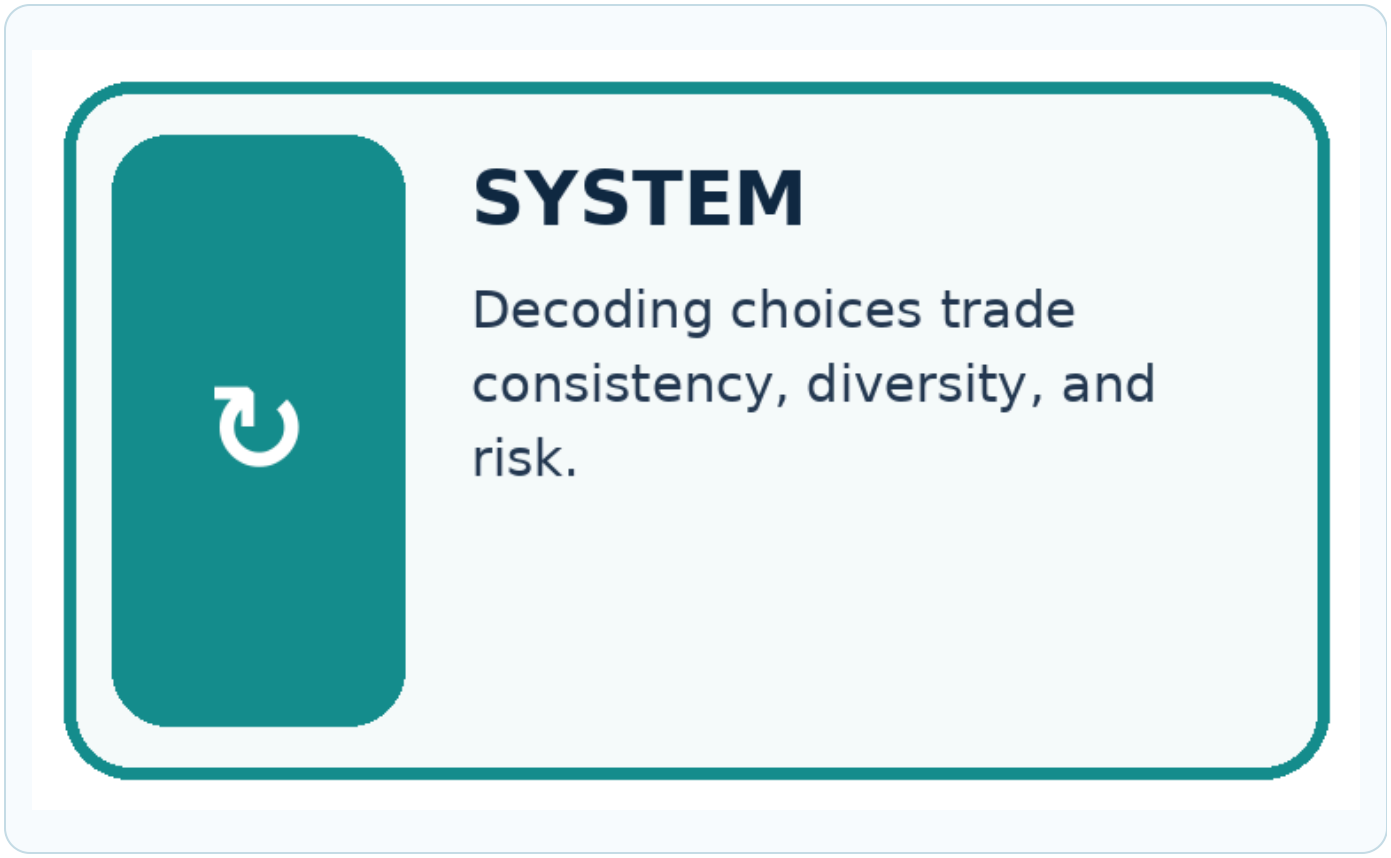
VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Compare low- and high-variability story continuations.

↓


SYSTEM
Decoding choices trade consistency, diversity, and risk.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

For strictly positive probabilities p_i and **temperature** T>0, define q_i = p_i^(1/T) / sum_j p_j^(1/T). This classroom formula is equivalent to scaling logits before a **softmax** when p came from those logits.

We use it directly on the supplied table. At T=1 the distribution is unchanged. At T=0.5, square each probability: 0.36,0.09,0.01; total 0.46. The adjusted distribution is red 18/23, blue 9/46, green 1/46.

Token	T=1 probability	T=0.5 probability
red	0.6	18/23 ≈ 0.782609
blue	0.3	9/46 ≈ 0.195652
green	0.1	1/46 ≈ 0.021739

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
T = 1	Intervals 0 to .6 to .9 to 1	Draw .7 selects blue
T = 0.5	First boundary 18/23	Draw .7 selects red
Same recorded draw	Changed distribution	Different output possible

Temperature changes probability concentration; the same draw can fall in a different interval.

Apply the connected idea: A small attention calculation

Item	Key, width 1	Value, width 1
A	[0]	[2]
B	[ln(3)]	[6]

Take q=[1] and d=1. The scale sqrt(1) is 1, so **score**s are 0 and ln(3). ln(3) is the number whose natural exponential is 3, approximately 1.098612. Exponentials have ratio 1:3, hence weights 1/4 and 3/4.

Output is (1/4)*2 + (3/4)*6 = 5. Using maximum subtraction gives exponentials 1/3 and 1 and the same normalised weights.

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
q=[1], keys 0 and ln(3)	Scaled scores 0, ln(3)	Softmax weights .25, .75
Values 2 and 6	Weighted sum .25*2 + .75*6	Output 5
Mask excludes B	Only A permitted	Weight 1 and output 2

Scores, normalised weights and value combinations are separate stages.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

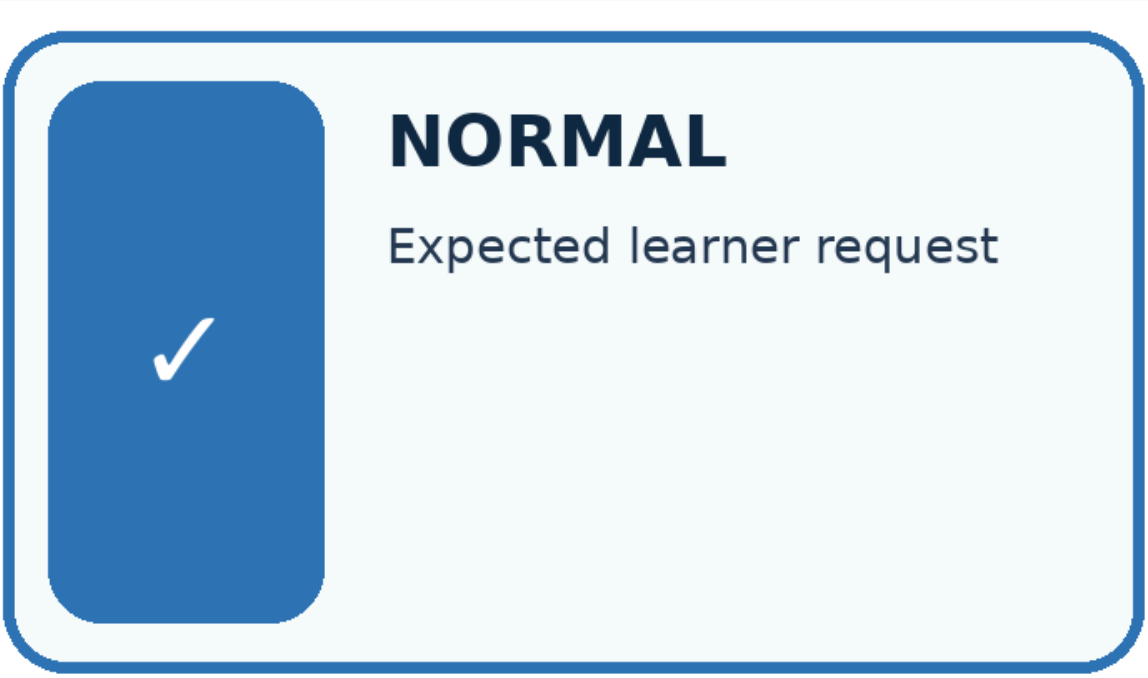
> Compare the supplied cases for Sampling and Variability. Identify the detail that changes the conclusion.

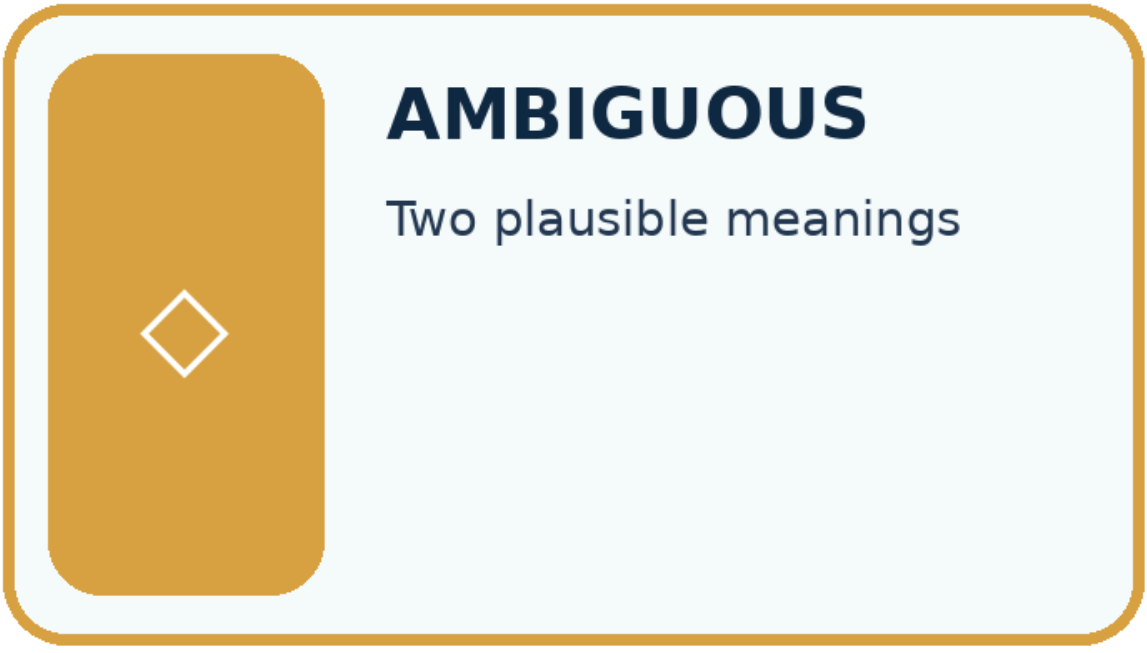
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Set the draw to 0.97, keeping both distributions.

Predicted result

Prediction: Both **temperature**s select B.

Why the result follows

Reason: 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Sampling and Variability to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

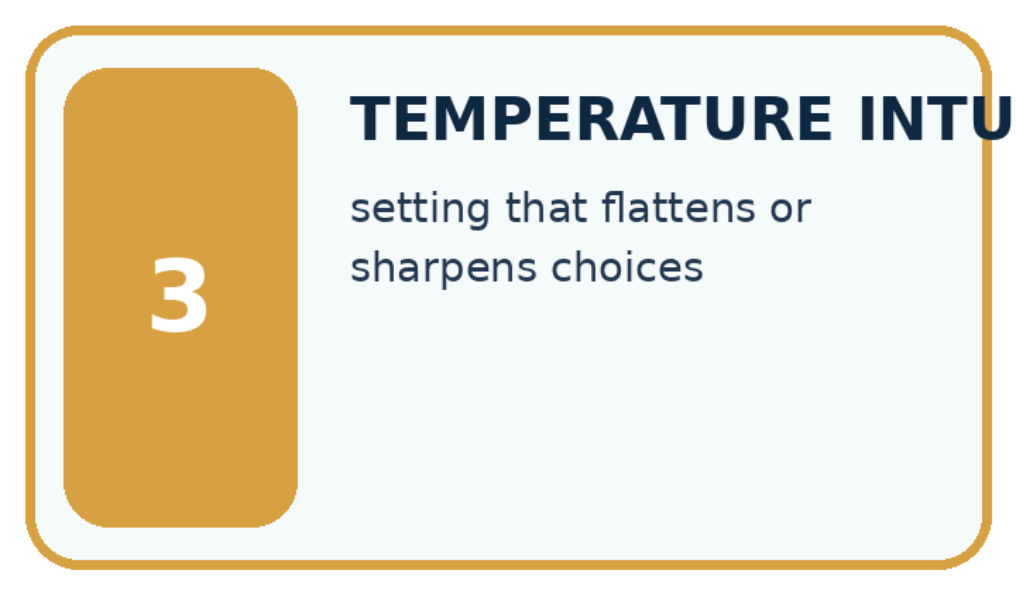
DECODING

method for choosing output tokens

2

VARIABILITY

how much repeated outputs differ





Word	My own example	Not this / boundary
decoding	Selecting output tokens from a model's scores using a specified procedure.	
variability	Differences observed across repeated runs, samples or conditions.	

Word	My own example	Not this / boundary
temperature intuition	A rough understanding of how temperature rescales token scores: higher values commonly spread choices more, without guaranteeing creativity or truth.	
seeded test	A test containing a deliberately planted, known condition or error to check whether it is handled.	

Words used in the closer explanation

Word	Meaning
Sampling	Selecting an alternative using a specified random procedure
Greedy choice	Selecting an alternative with largest next-token probability under a tie rule
Temperature	A positive parameter that changes the concentration of a distribution under a stated transformation
Cumulative interval	A range of draws assigned to an alternative using accumulated probability mass

Terms for A small attention calculation

Term	Meaning
Scaled dot product	A query-key dot product divided by the square root of key width
Softmax	Exponentiating scores and normalising their sum to obtain weights
Score	An unnormalised compatibility value before softmax
Numerical stability	An implementation choice that avoids avoidable arithmetic overflow or loss

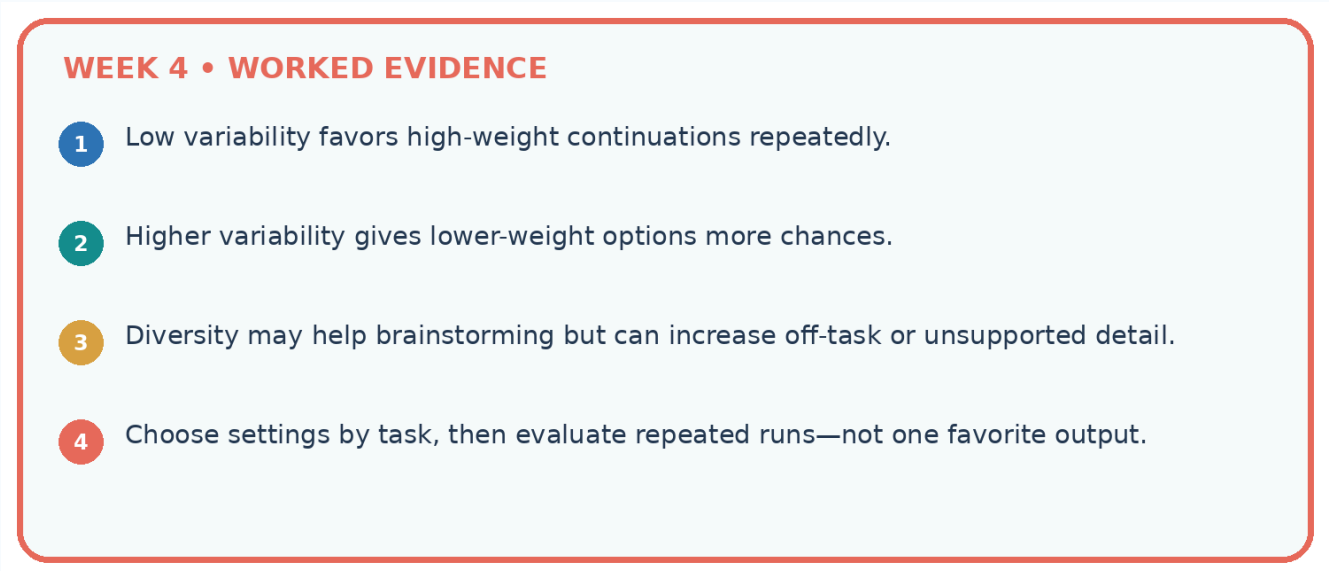
Follow a worked example

> Follow the worked trace for Sampling and Variability, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Low **variability** favors high-weight continuations repeatedly.

2. Higher **variability** gives lower-weight options more chances.

3. Diversity may help brainstorming but can increase off-task or unsupported detail.

4. Choose settings by task, then evaluate repeated runs—not one favorite **output**.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Worked explanation · Sampling and Variability

The lower positive **temperature** concentrates this table on its largest entry. For T=2, use square roots before normalising; this flattens this unequal distribution.

These changes leave the order of positive probabilities unchanged, so greedy still selects red. They change the **sampling** distribution, not the factual **evidence** behind a generated claim.

Worked explanation · Sampling and Variability

Our formula excludes T=0; dividing by zero is undefined. Some systems offer a zero-**temperature** setting as a separate greedy policy, but that is not a value to substitute into this formula.

We also avoid claiming that any one setting removes all nondeterminism in an unspecified system.

Apply the connected idea: A small attention calculation

Worked explanation · Sampling and Variability

For q=[1,1], kA=[1,0] and kB=[0,1], both dot products are 1. Divide by sqrt(2); equal **score**s give equal weights 1/2 regardless of their common value. With values [2,0] and [0,4], the **output** is [1,2].

Do not omit the scale merely because it cancels in this tied example; unequal **score**s generally change concentration when scaled.

Worked explanation · Sampling and Variability

Apply masks before normalising over permitted **score**s. In the width-one example, excluding B leaves weight 1 for A and **output** 2. A key with the wrong width violates the dot-product contract.

Values must share a width with one another, but their width need not equal the query/key width. An all-excluded mask is handled explicitly instead of taking a maximum of an empty list.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sampling and Variability

A fictional two-token distribution illustrates **variability** in generation.

The facts we will use

> **Supplied facts**
>
> At T=1, A=0.8 and B=0.2. For positive T use p^(1/T), then normalise.

> **Supplied facts**
>
> Use token order A,B and **cumulative interval**s [0,boundary),[boundary,1). Draw u=0.85.

AI ACADEMY / AI-3/4

Temperature-scaled toy probabilities

	T=1	T=0.5
A	0.8	0.9411764705882353
B	0.2	0.058823529411764705

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/4

Sampling and Variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Original distribution

A boundary 0.8; u=0.85 selects B.



2. T=0.5 weights

Square probabilities:0.64 and 0.04; total 0.68.



3. Normalise

A=16/17≈0.9412; B=1/17≈0.0588.



4. Same draw

0.85 is below 16/17, so T=0.5 selects A.

CHANGE ONE THING

Set the draw to 0.97, keeping both distributions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Original distribution	A boundary 0.8; u=0.85 selects B.
2. T=0.5 weights	Square probabilities:0.64 and 0.04; total 0.68.
3. Normalise	A=16/17≈0.9412; B=1/17≈0.0588.
4. Same draw	0.85 is below 16/17, so T=0.5 selects A.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Lower positive **temperature** concentrates this unequal distribution on its larger entry. It changes **sampling** probabilities, not **evidence** for whether A or B is true.

Using the same draw isolates how the distribution changes the decision. This formula is a toy probability-space illustration corresponding to a **temperature**-scaled model distribution.

What this example does not establish

Keep the stated limit

Two tokens and one draw do not predict every **output**; **temperature** alone does not solve hallucinations.

Keep the stated limit

Read the labelled picture: Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?

Keep the stated limit

Change one thing: Set the draw to 0.97, keeping both distributions. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Sampling and Variability claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Higher variability makes a model more knowledgeable. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

A fair **temperature** comparison fixes the prefix, model distribution, candidate order and draw policy. Record multiple outcomes and evaluate task criteria separately.

If the aim is creativity in fiction, variation may be useful; if the aim is reporting a sensor state, every factual claim still needs **evidence**.

Never call a probability-concentration change an accuracy improvement without an appropriate evaluation.

Exit: a team lowers temperature and claims hallucinations are impossible. State what was changed and what evidence that claim still lacks.

Apply the connected idea: A small attention calculation

Check the claim against the facts

Check query/key dimensions, scale, permitted positions, finite **score**s, stable normalisation, weight sum and value widths in that order. A zero **output** can result from values cancelling; it does not imply zero attention weights.

Record enough intermediate values for someone else to reproduce the trace.

Exit: two equal scores yield weights 1/2,1/2 and values 4,-4 yield output 0. Explain why zero output does not mean neither position was used.

STUDENT LESSON · 8 OF 18

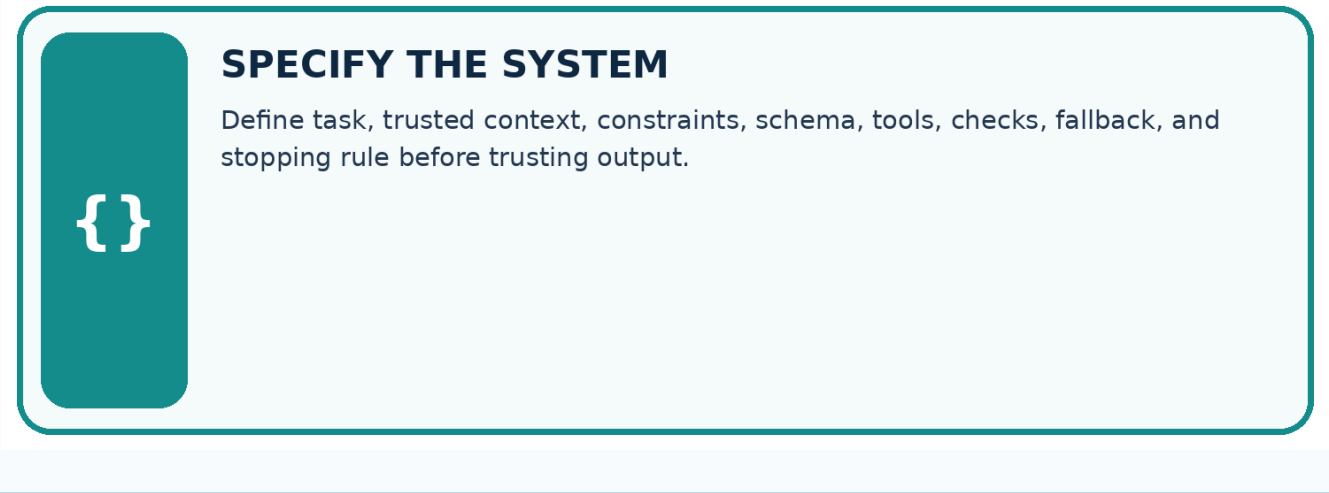
Practise together

Practise Sampling and Variability with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Supported draws 0.2,0.6,0.9 at T=1 select red, blue, green. At T=0.5, 0.2 and 0.6 fall below 18/23 and select red; 0.9 lies between 18/23 and 45/46 and selects blue.

Keep exact fractions for boundaries, because rounding can misclassify a draw near an endpoint.

Guided: trace those three draws under both distributions and explain why the result is a procedure trace, not a frequency estimate from a large experiment.

Apply the connected idea: A small attention calculation

Supported **score**s 1000 and 1000+ln(3) would overflow an ordinary direct exponential calculation in many floating-point implementations. Maximum subtraction gives -ln(3) and 0, producing weights 1/4 and 3/4.

Their common shift does not change the relative weights. This is a stability **mechanism**, not **evidence** about source correctness.

Guided: calculate the shifted scores and weights, then the output for values 4 and 8.

Practise with less help: Compute the adjusted table

Use workbook W2 for the connected example "**Sampling** and **temperature**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Sampling** and **temperature**. Proposed explanation: "**Sampling** means equal chances". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: A small attention calculation

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: **Score**s already are probabilities

Explain the picture

> Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

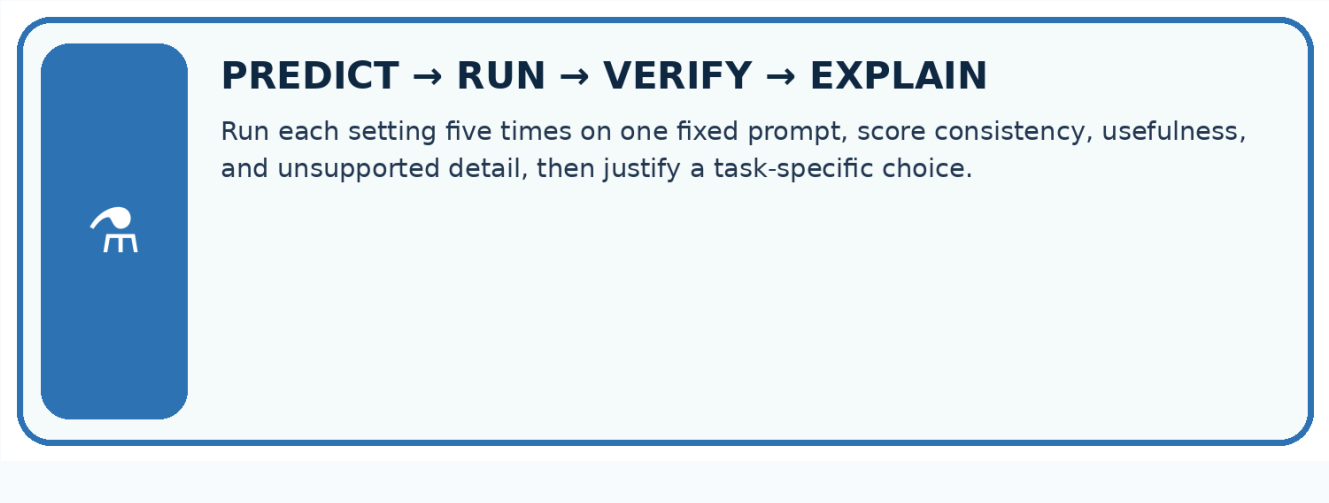
Run the investigation

> Investigate Sampling and Variability. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Run each setting five times on one fixed prompt, **score** consistency, usefulness, and unsupported detail, then justify a task-specific choice.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

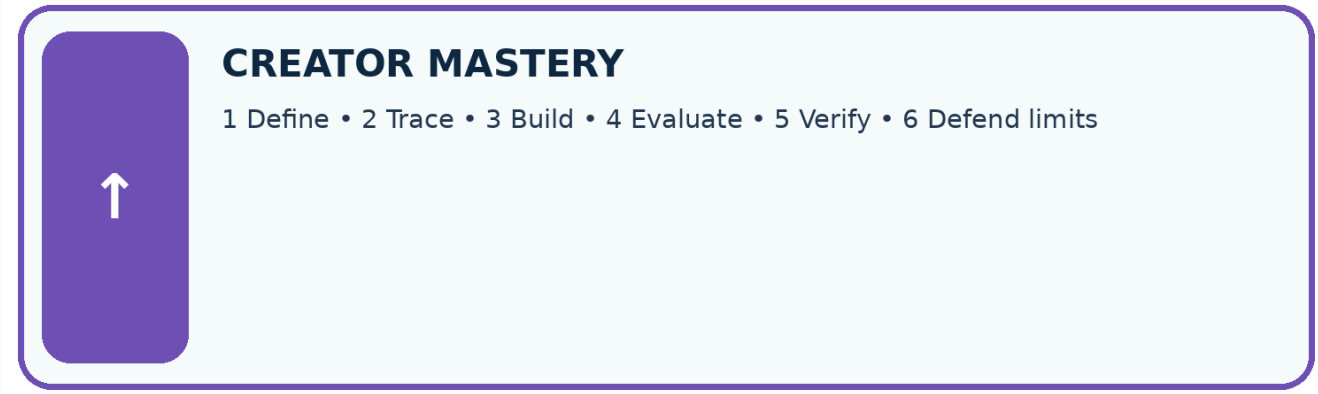
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Sampling and Variability. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.

VERIFIED STUDY BUDDY

Use low variability for structured study answers and allow more diversity only for labeled brainstorming.

THIS WEEK'S CHECKPOINT Use low **variability** for structured study answers and allow more diversity only for labeled brainstorming.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

Keep Study Buddy generation settings fixed during prompt comparisons and check **evidence** regardless of **variability**.

STUDENT LESSON · 12 OF 18

Show what you know

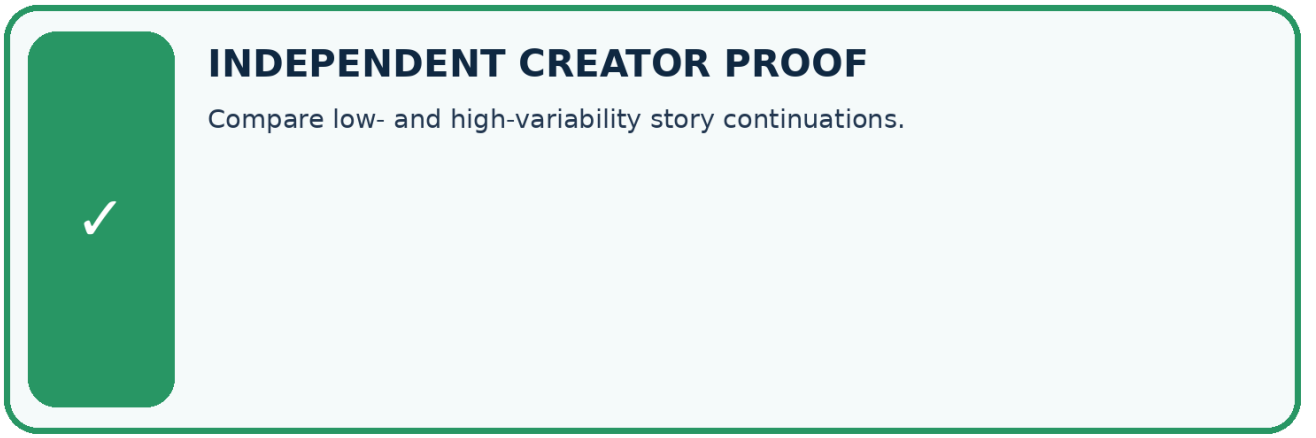
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Compare low- and high-**variability** story continuations.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Sampling and Variability case for supported practice. Repeated examples do not become fresh assessments.

Week 4 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

One prompt produces three different valid quiz questions.

The question

> Does **variability** mean failure?

Worked explanation

No. **Sampling** can vary **output**s. **Score** correctness, relevance and format over repeated trials; consistency differs from quality.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Variability and quality are separate

Evidence or stage	Value or result
Trial A	Different valid quiz question
Trial B	Another valid quiz question
Trial C	Check correctness, relevance and format

One prompt produces three different valid quiz questions.

> Does **variability** mean failure?

No. **Sampling** can vary **output**s. **Score** correctness, relevance and format over repeated trials; consistency differs from quality.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Sample a new table independently

Probabilities A=0.5,B=0.3,C=0.2 in that order. Compute T=0.5 probabilities and cumulative boundaries. Trace u=0.5 and u=0.9 under T=1 and T=0.5. Explain the limit of the comparison.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 4 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Prompt A is run three times and returns three different stories. Does this alone prove the prompt changed? Describe a fair comparison of two prompts.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Use low **variability** for structured study answers and allow more diversity only for labeled brainstorming.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Prompt A gives three identical wrong answers. Prompt B gives two correct answers and one wrong answer. Compare consistency and correctness without declaring either generally superior.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Sampling and Variability, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional two-token distribution illustrates **variability** in generation.

By the end, I can	How I will show it
Sample from a distribution using explicit cumulative intervals	Explain the deciding step, show a trace or test, and state one limit.
Calculate temperature-adjusted probabilities for a toy example	Explain the deciding step, show a trace or test, and state one limit.
Explain why random variation and factual reliability need separate evaluation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Sampling and Variability without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Sample from a distribution using explicit **cumulative interval**s

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional two-token distribution illustrates **variability** in generation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? 0.85 is below 16/17, so T=0.5 selects A.

1. Normalise

2. Original distribution

3. T=0.5 weights

4. Same draw

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage

that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Same draw 0.85 is below 16/17, so T=0.5 selects A. Lower positive **temperature** concentrates this unequal distribution on its larger entry.

It changes **sampling** probabilities, not **evidence** for whether A or B is true. Using the same draw isolates how the distribution changes the decision.

This formula is a toy probability-space illustration corresponding to a **temperature**-scaled model distribution.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Set the draw to 0.97, keeping both distributions. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Both **temperature**s select B.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Both **temperature**s select B. 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

Repair a missing building block

Review Tokens and Context (ai-3/3). Goal: Account for fixed text, **evidence** and reserved **output** under a toy budget. Short practice: Can A alone justify answering 20 seats after the exception is known?.

Return to this week and explain what you changed.

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

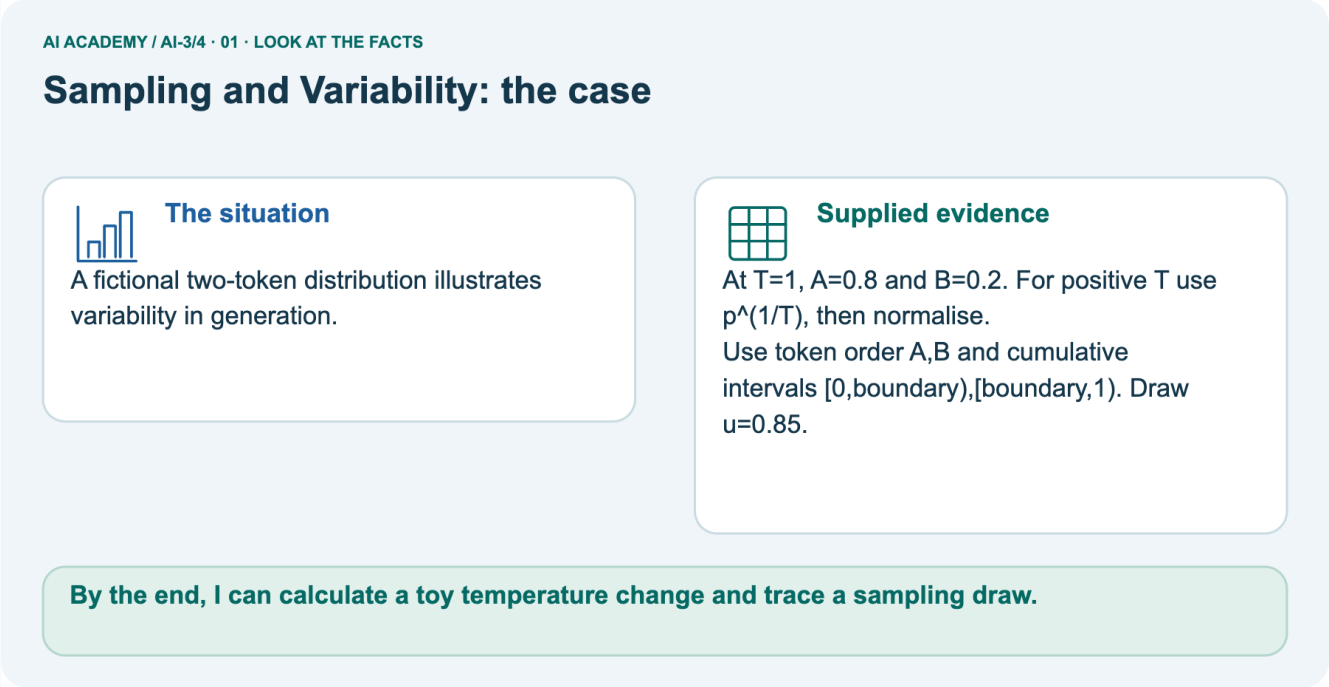
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Sampling and Variability graphic using its labels, deciding evidence and relevant limitation.

01. Sampling and Variability: the case



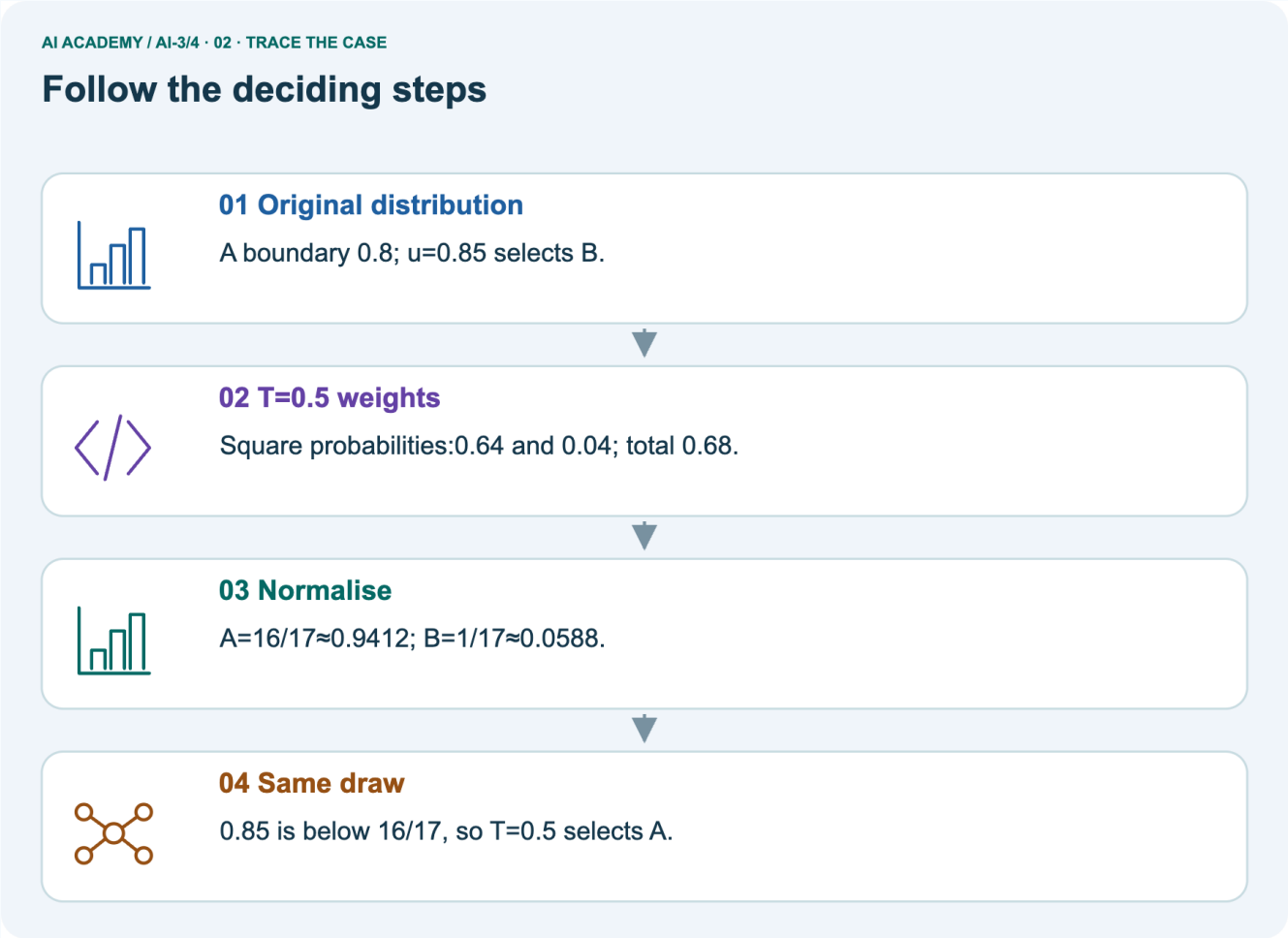
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional two-token distribution illustrates **variability** in generation. Supplied facts: At T=1, A=0.8 and B=0.2. For positive T use p^(1/T), then normalise.

Use token order A,B and **cumulative interval**s [0,boundary),[boundary,1). Draw u=0.85.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

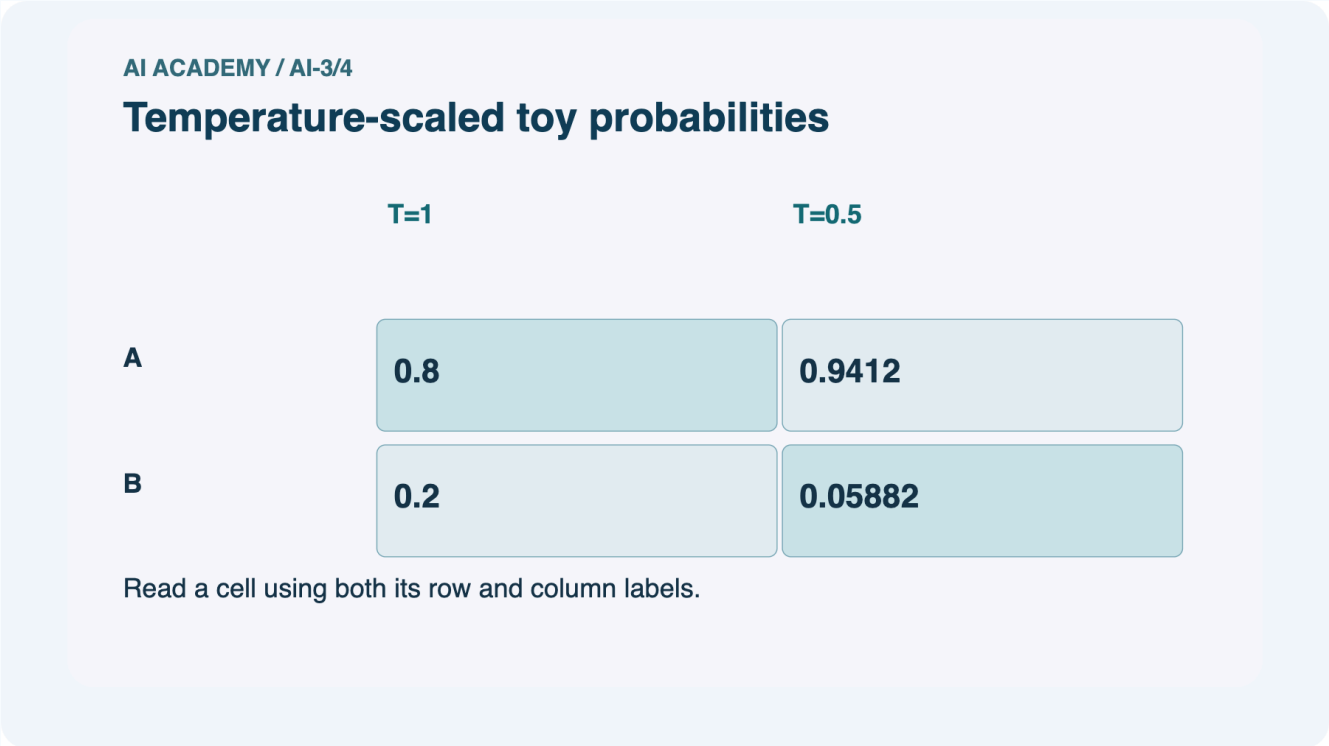


Follow the labelled picture with the matching explanation. Open the vector diagram

Original distribution: A boundary 0.8; u=0.85 selects B. → T=0.5 weights: Square probabilities:0.64 and 0.04; total 0.68. → Normalise: A=16/17≈0.9412; B=1/17≈0.0588. → Same draw: 0.85 is below 16/17, so T=0.5 selects A.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Temperature-scaled toy probabilities

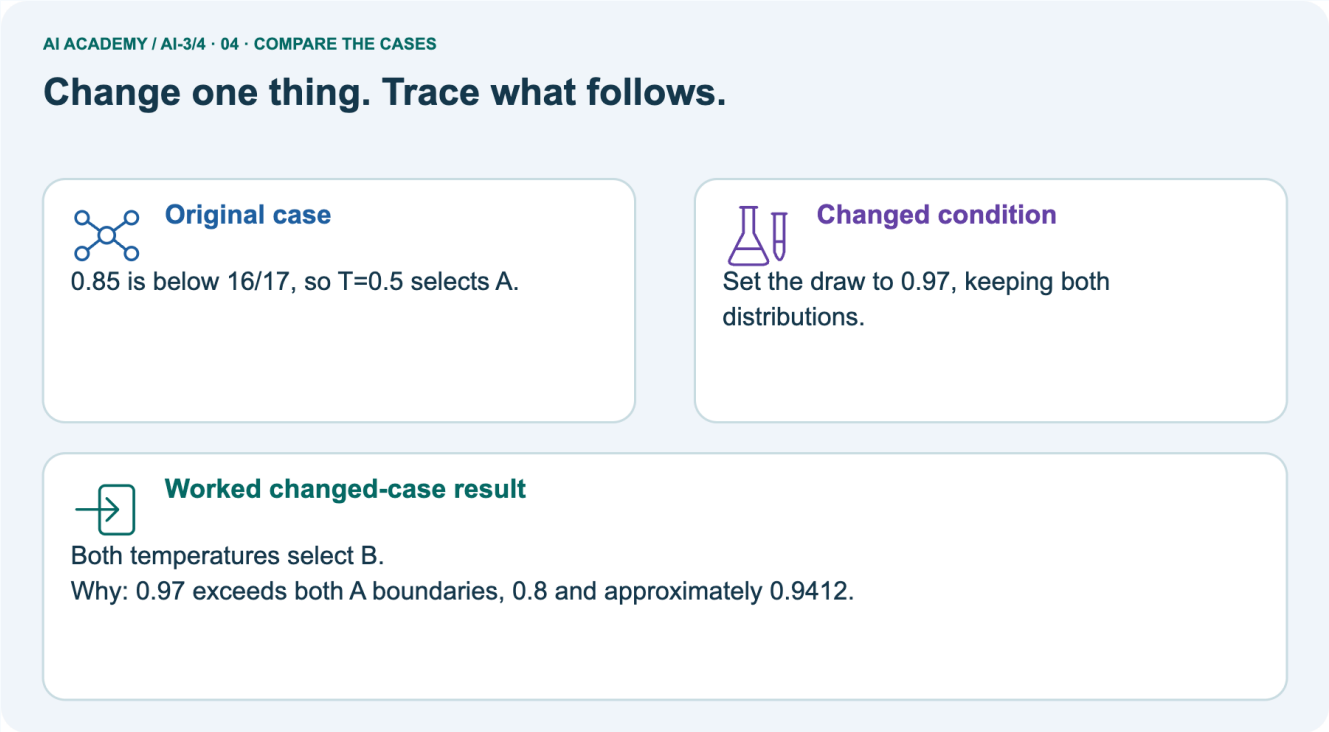


Follow the labelled picture with the matching explanation. Open the vector diagram

Temperature-scaled toy probabilities. A: T=1 = 0.8, T=0.5 = 0.9411764705882353. B: T=1 = 0.2, T=0.5 = 0.058823529411764705.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



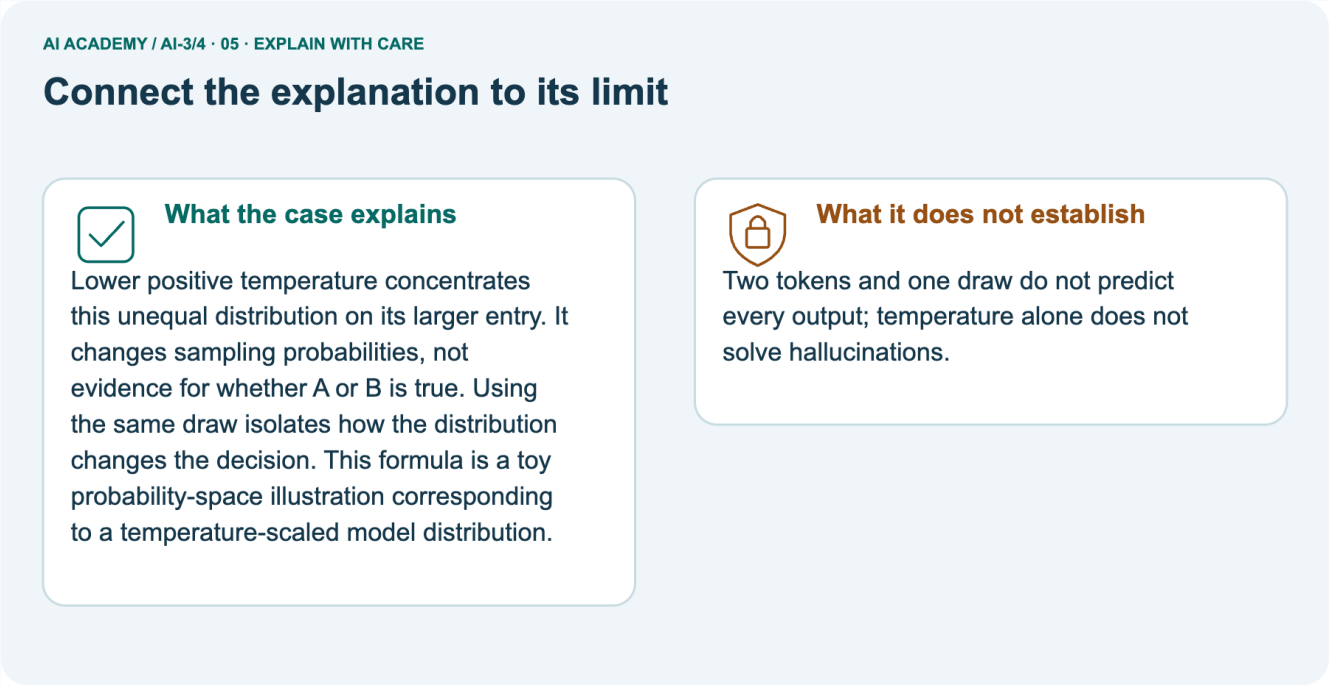
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 0.85 is below 16/17, so T=0.5 selects A. Changed condition: Set the draw to 0.97, keeping both distributions. Expected result: Both **temperature**s select B.

Reason: 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

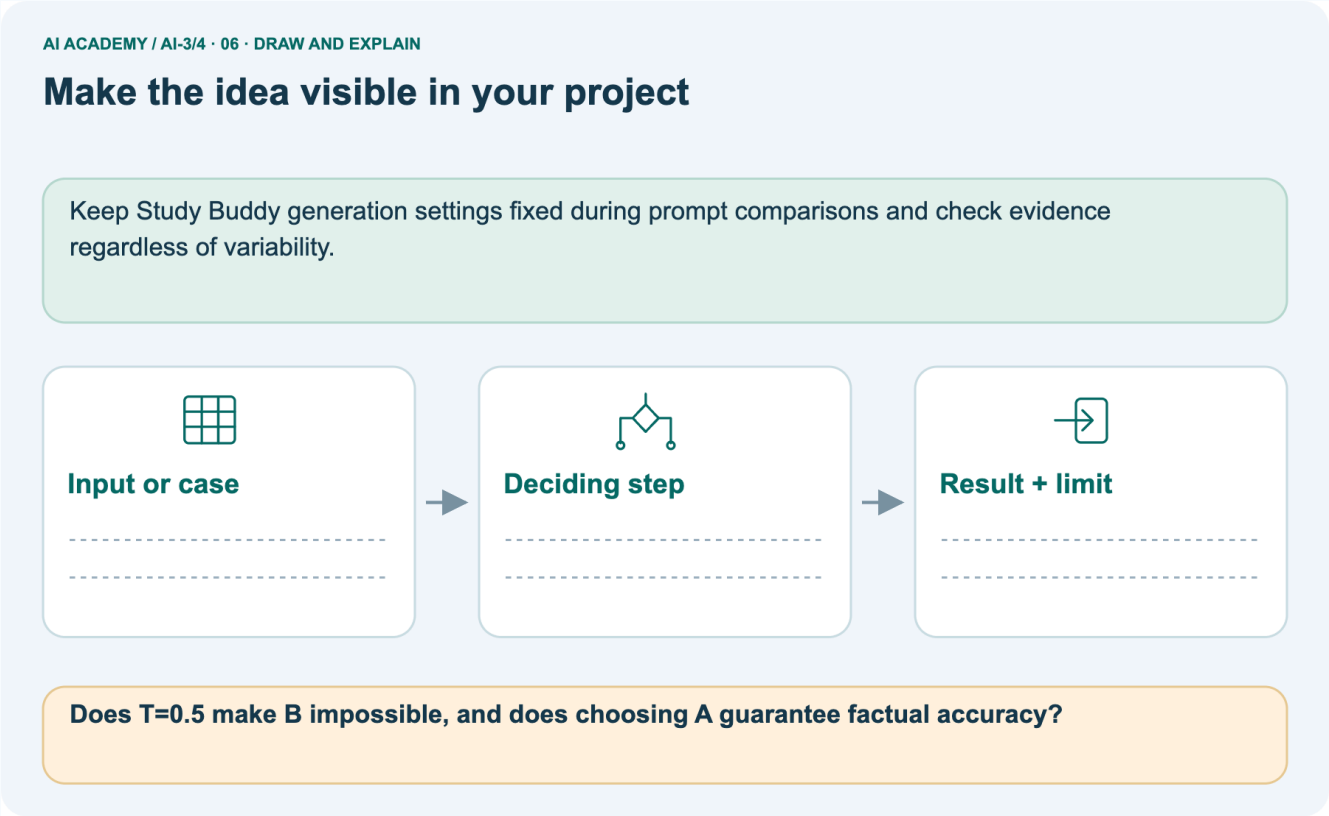
Lower positive **temperature** concentrates this unequal distribution on its larger entry. It changes **sampling** probabilities, not **evidence** for whether A or B is true.

Using the same draw isolates how the distribution changes the decision. This formula is a toy probability-space illustration corresponding to a **temperature**-scaled model distribution.

Limit: Two tokens and one draw do not predict every **output**; **temperature** alone does not solve hallucinations.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep Study Buddy generation settings fixed during prompt comparisons and check **evidence** regardless of **variability**. Draw your **input**, deciding step, result and limit.

Supported question: Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.